Screening Tes

Community Medicine (PSM) | 3rd Year

Hi Sammy!
Screening = numericals + theory gold mine!
Must-know for PSM!

Screening Tests | PSM | 3rd Year | MBBS

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Wilson & Jungner criteria
= asked EVERY year!
Learn all 10 points!

DEFINITION

"The presumptive identification of unrecognized disease by tests which can be applied rapidly & economically." — Park

Screening Test vs Diagnostic Test

Feature	Screening Test	Diagnostic Test
Purpose	Identify probable cases	Confirm diagnosis
Applied to	Apparently healthy people	Symptomatic / suspected cases
Cost	Low (mass application)	High (individual)
Speed	Rapid	Time-consuming
Sensitivity	High (few false negatives)	May vary
Specificity	Moderate	High (few false positives)
Follow-up	Needs confirmation	Definitive result

Types of Screening

Type	Description
Mass screening	Applied to entire population irrespective of risk
Selective/Targeted	Applied to high-risk groups only (e.g., contacts of TB)
Multiphasic	Multiple screening tests applied simultaneously
Opportunistic	Screening when patient visits health facility for other reasons
Case finding	Screening by clinician during routine clinical care

Wilson + Jungner Criteria (WHO 1968)

1. Important health problem
2. Accepted treatment available for recognized disease
3. Facilities for diagnosis & treatment available
4. Recognizable latent or early symptomatic stage
5. Suitable test or examination exists
6. Test acceptable to the population
7. Natural history of disease adequately understood
8. Agreed policy on whom to treat as patients
9. Cost of case-finding economically balanced
10. Case-finding should be a continuing process

The 2x2 table is THE most important thing! Numericals come from here!

The 2x2 Table (Foundation of Everything!)

	Disease + (D+)	Disease - (D-)	Total
Test + (T+)	True Positive (a)	False Positive (b)	a + b
Test - (T-)	False Negative (c)	True Negative (d)	c + d
Total	a + c	b + d	a+b+c+d

Key Formulas (Must Memorize!)

Measure	Formula	Meaning
Sensitivity	a / (a + c) x 100	How well test detects disease
Specificity	d / (b + d) x 100	How well test rules out disease
PPV (+ve pred. val.)	a / (a + b) x 100	If +ve, probability of disease
NPV (-ve pred. val.)	d / (c + d) x 100	If -ve, probability of no disease
Prevalence	(a + c) / (a+b+c+d) x 100	Disease frequency in population
Accuracy	(a + d) / (a+b+c+d) x 100	Overall correct results

★ Effect of Prevalence on PPV & NPV:

Prevalence INCREASES -> PPV INCREASES, NPV DECREASES

Prevalence DECREASES -> PPV DECREASES, NPV INCREASES

Clinical Mnemonics (SnNOut / SpPIn)

SnNOut

Sensitive test, Negative result

= Rules OUT disease
(Few false negatives)

SpPIn

Specific test, Positive result

= Rules IN disease
(Few false positives)

★ Remember This!

Sensitivity & Specificity are INVERSELY related (trade-off).

Sensitivity & Specificity are FIXED for a test (independent of prevalence).

ROC curves & biases are tricky but very important for exams!

ROC Curve (Receiver Operating Characteristic)

- Plot of Sensitivity (Y-axis) vs 1 - Specificity (X-axis)
- Area Under Curve (AUC) = overall test accuracy
- AUC = 1.0 = perfect test | AUC = 0.5 = useless test (diagonal line)
- Used to: (1) Choose optimal cut-off point (2) Compare different tests
- Curve closer to TOP-LEFT corner = better test performance

Likelihood Ratios (LR)

Measure	Formula	Interpretation
LR+ (Positive)	Sensitivity / (1 - Specificity)	Higher = better at ruling IN
LR- (Negative)	(1 - Sensitivity) / Specificity	Lower = better at ruling OUT
LR+ > 10	Strong evidence	Almost confirms disease
LR- < 0.1	Strong evidence	Almost excludes disease

Biases in Screening Programmes

Bias	Explanation
Lead time bias	Early detection makes survival APPEAR longer, but death unchanged
Length time bias	Screening detects slow-growing (indolent) disease preferentially
Selection/Volunteer	Health-conscious people volunteer -> better outcomes anyway
Overdiagnosis bias	Detecting disease that would never cause symptoms/death

Lead Time Bias Example

Without screening: diagnosed at 60, dies at 65 = 5 yr survival.

With screening: detected at 55, STILL dies at 65 = 10 yr survival.

~~Patient didn't live longer — just diagnosed earlier! Don't be fooled!~~

Ideal Screening Test Characteristics

1. High sensitivity (detect maximum cases) & high specificity
2. Simple, rapid, inexpensive, and safe to perform
3. Acceptable to the population (non-invasive, painless)
4. Reliable (reproducible) and valid (accurate)

National programmes with screening = asked in LAQs every year!

National Programmes with Screening Component

Programme	Screening Details
NPCBVI (1976)	Cataract screening >50 yrs, school eye screening
RNTCP/NTEP	Screening cough >2 weeks, sputum microscopy (ZN stain)
NPCDCS (2010)	Opportunistic screening for DM, HTN, oral/breast/cervical Ca
NACP	Screening of high-risk groups for HIV (ICTC/PPTCT)
NVBDCP	Fever screening for malaria (microscopy + RDT)
NMHP (1982)	Screening for mental illness at PHC level
UIP	Antenatal screening (Hb, blood group, HIV, VDRL, urine)

Cancer Screening (NPCDCS) — High Yield!

Cancer	Method	Target Population
Oral cancer	Visual examination of oral cavity	All males >30 yrs (tobacco users)
Breast cancer	Clinical Breast Examination (CBE)	All females >30 yrs
Cervical cancer	VIA (Visual Inspection w/ Acetic acid)	All females 30-65 yrs

Important Screening Tests (One-Liners)

Disease	Screening Test
Diabetes mellitus	Fasting blood glucose / OGTT
Hypertension	Blood pressure measurement
Cervical cancer	Pap smear / VIA
Breast cancer	Mammography / CBE
Congenital hypothyroidism	TSH (neonatal screening)
PKU (Phenylketonuria)	Guthrie test (heel prick)
Sickle cell disease	Sickling test / HPLC
Pulmonary TB	Sputum microscopy / CBNAAT
HIV	ELISA (screening) -> Western Blot (confirmatory)
Glaucoma	Tonometry (IOP measurement)

★ HIV Testing Strategy

ELISA = high SENSITIVITY (screening) -> SnNOut principle.

Western Blot = high SPECIFICITY (confirmatory) -> SpPIn principle.

Exam Strategy + High Yield

Screening = guaranteed
15+ marks in PSM!
You've GOT this Sammy!

Most Repeated LAQs (10-12 marks)

- 1. Define screening. Describe Wilson & Jungner criteria for screening
- 2. Explain sensitivity, specificity, PPV, NPV with 2x2 table
- 3. Numerical: Calculate sensitivity, specificity, PPV, NPV from given data
- 4. Effect of prevalence on PPV and NPV. Explain with examples
- 5. Describe national screening programmes in India with examples

Most Repeated SAQs (5-6 marks)

- 1. Differentiate screening test vs diagnostic test
- 2. Define and calculate sensitivity and specificity
- 3. SnNOut and SpPIn — explain with examples
- 4. ROC curve — definition and uses
- 5. Lead time bias and length time bias
- 6. Likelihood ratios — positive and negative

Numerical Problem Pattern (MUST Practice!)

1. **Step 1** — Draw the 2x2 table from given data
2. **Step 2** — Identify a, b, c, d from the question
3. **Step 3** — Apply formulas: Sens = a/(a+c), Spec = d/(b+d)
4. **Step 4** — Calculate PPV = a/(a+b), NPV = d/(c+d)
5. **Step 5** — Comment on prevalence effect if asked
6. **Step 6** — Calculate LR+ and LR- if needed

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Screening test with highest sensitivity	Used to RULE OUT (SnNOut)
Screening test with highest specificity	Used to RULE IN (SpPIn)
PPV depends on	Prevalence of disease
Gold standard for cervical cancer screening	Pap smear (now HPV DNA test)
Screening test for neonatal hypothyroidism	TSH (heel prick blood)
Wilson & Jungner criteria (year)	WHO, 1968 (10 criteria)
AUC = 0.5 on ROC curve means	Test is useless (no better than chance)
Lead time bias affects	Survival time (appears longer)
ELISA is used as screening for	HIV (high sensitivity)
Guthrie test screens for	Phenylketonuria (PKU)

♥ **Screening saves lives by catching disease early, Sammy!**
Master these formulas and you'll ace every numerical. Pihu believes in you!